

NOTES ON MARKOV CHAINS

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1. INTRODUCTION

These notes provide a fairly complete account of the Markov convergence theorem, which is the basis for Markov Chain Monte Carlo methods. All of the proofs are elementary (not using any advanced mathematics), but not particularly easy (since the arguments are fairly deep). The setup is not the most general (I assume a finite state space), but I've tried to use arguments that generalize without too much modification. One proof (involving some number theory) is skipped, though the result seems intuitively plausible. Otherwise, everything is proven from first principles.

On a first reading, it is best to focus on concepts. There are a small number of definitions (the Markov property, homogeneity, communicating states, irreducibility, aperiodicity, stationary distributions), each of which is easily comprehended. Then there are about five important results:

- A Markov chain is characterized by its initial distribution and a matrix of transition probabilities.
- Two states communicate if there is a path with strictly positive transition probabilities connecting the two states.
- If the chain is irreducible and aperiodic, it has a unique stationary distribution giving the long run proportion of time occupied in each state.
- If the chain is irreducible and aperiodic, the chain converges (in a sense made precise below) to its stationary distribution.

We save the concept of reversability for the next session, which provides an easy way to construct a Markov chain whose stationary distribution is the posterior distribution for inference in a Bayesian model (the Metropolis-Hastings algorithm, which includes the Gibbs sampler as a special case).

Literature. The classic exposition is in Chapter XV of Feller, *An Introduction to Probability Theory and Its Applications*, volume 1. An accessible introduction to the modern theory of Markov chains is Levin *et al.*, *Markov Chains and Mixing Times*, which these notes follow closely.

2. DEFINITIONS AND EXAMPLES

A *stochastic process* $\{X_t\}$ is a collection of random variables indexed by time t . Time can be either discrete (*e.g.*, when t takes integer values) or continuous (*e.g.*, when t is real-valued). For present purposes, we will stick to discrete-time stochastic processes with time starting at $t = 0$. That is, the stochastic process will be a sequence of random variables $X_0, X_1, \dots$.

The *state space* $\mathcal{X}$ of the stochastic process is the set of values that the random variables take. In these notes, the state space will be finite or countable, though statistical applications often involve uncountable state spaces (*e.g.*, continuous random variables), and, aside from a lot of technical

details, similar results hold in general state spaces. Intuition is easiest for finite state spaces, so we will usually stick to this case. Without loss of generality, the states will be denoted by integers starting at either zero or one, depending upon which is most convenient.

A *finite Markov Chain* is a sequence of random variables $X_0, X_1, \dots$, such that

$$\mathbb{P}\{X_{n+1} = x_{n+1} | X_0 = x_0, \dots, X_{t-1} = x_n\} = \mathbb{P}\{X_{n+1} = x_{n+1} | X_n = x_n\}$$

for $n = 0, 1, \dots$, and any values $x_0, \dots, x_{n+1}$ on the state space $\mathcal{X}$. That is, the next value X_{n+1} of the stochastic process is independent of the past values $X_0, \dots, X_{n-1}$ conditional upon the present X_n . This condition is known as the *Markov property*.

Example 1. Suppose you enter a casino with X_0 dollars and place a sequence of bets, each for one dollar, where you win with probability $1/2$ and lose with probability $1/2$. Let X_t denote your wealth after the t bet. You continue betting until you run out of money ($X_t = 0$) or you bankrupt the casino ($X_t = K$ if the casino has capital $K - X_0$). This problem is known as *Gambler's ruin*, since that's how it usually ends. The state space is $\mathcal{X} = \{0, 1, \dots, K\}$.

The probability of a sample path $(x_0, \dots, x_n)$ is given by

$$\begin{aligned} \mathbb{P}\{X_0 = x_0, \dots, X_n = x_n\} &= \mathbb{P}\{X_0 = x_0\} \prod_{t=1}^n \mathbb{P}\{X_t = x_t | X_0 = x_0, \dots, X_{t-1} = x_{t-1}\} \\ &= \mathbb{P}\{X_0 = x_0\} \prod_{t=1}^n \mathbb{P}\{X_t = x_t | X_{t-1} = x_{t-1}\}. \end{aligned}$$

Define the *transition probabilities*

$$p_{ij} = \mathbb{P}\{X_t = j | X_{t-1} = i\} \quad (t = 1, 2, \dots; i, j \in \mathcal{X})$$

We are (implicitly) assuming *homogeneity*: the transition probabilities do not depend on t . We can arrange the transition probabilities into a matrix $\mathbf{P}$, e.g.,

$$\mathbf{P} = \begin{pmatrix} p_{11} & \cdots & p_{1k} \\ \vdots & & \vdots \\ p_{k1} & \cdots & p_{kk} \end{pmatrix}$$

if $\mathcal{X} = \{1, \dots, P\}$.

Example 2. Continuing the previous example, suppose $X_0 = 1$ and $K = 2$. Then $\mathcal{X} = \{0, 1, 2, 3\}$ and $\mathbf{P}$ is the 4×4 matrix

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The states 0 and 3 are called *absorbing states*: once you reach them, you never leave. One of these is much preferable than the other, depending upon your perspective.

Since the rows of $\mathbf{P}$ represent a conditional probability distribution, the elements are non-negative with row sums equal to one.

$$p_{ij} \geq 0 \quad \sum_{j \in \mathcal{X}} p_{ij} = 1$$

A matrix with these properties is said to be *stochastic*.

Let $\pi^{(0)}$ denote the *initial distribution*, i.e. the probability distribution of X_0 . $\pi^{(0)}$ will have as many elements as the state space $\mathcal{X}$. For example, if the state space is $\mathcal{X} = \{1, \dots, K\}$, then

$$\pi^{(0)} = \begin{pmatrix} \pi_1^{(0)} & \dots & \pi_K^{(0)} \end{pmatrix}$$

where

$$\pi_i^{(0)} = \mathbb{P}\{X_0 = i\} \quad (i \in \mathcal{X})$$

In general, let $\pi^{(t)}$ denote the (marginal) probability distribution of X_t , where

$$\pi^{(t)} = \begin{pmatrix} \pi_1^{(t)} & \dots & \pi_K^{(t)} \end{pmatrix}$$

where

$$\pi_i^{(t)} = \mathbb{P}\{X_t = i\} \quad (i \in \mathcal{X})$$

The distributions $\pi^{(t)}$ can be obtained from the initial distribution $\pi^{(0)}$ and the transition matrix $\mathbf{P}$ through matrix multiplication.

Proposition 1. $\pi^{(n)} = \pi^{(0)} \mathbf{P}^n$.

Proof. The proof is by induction on n . The case $n = 0$ is true by definition. By the Law of Total Probability,

$$\pi_j^{(n+1)} = \sum_{i \in \mathcal{X}} \mathbb{P}\{X_n = i\} \mathbb{P}\{X_{n+1} = j | X_n = i\} = \sum_{i \in \mathcal{X}} \pi_i^{(n)} p_{ij},$$

which is the dot product of $\pi^{(n)}$ and the j th column of $\mathbf{P}$. In matrix notation, $\pi^{(n+1)} = \pi^{(n)} \mathbf{P}$. Using the induction hypothesis, $\pi^{(n)} = \pi^{(0)} \mathbf{P}^n$, so $\pi^{(n+1)} = \pi^{(0)} \mathbf{P}^n \mathbf{P} = \pi^{(0)} \mathbf{P}^{n+1}$. $\square$

3. OVERVIEW OF THE PROBLEM

We are interested in the long-run behavior of Markov chains. Two examples, one good and the other pathological, illustrate the problem.

Example 3. Suppose we have the transition matrix

$$\mathbf{P} = \begin{pmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{pmatrix}$$

with initial distribution $\pi^{(0)} = (\pi^{(0)}, 1 - \pi^{(0)})$. Then

$$\pi^{(n+1)} = \frac{3}{4} \pi^{(n)} + \frac{1}{2} (1 - \pi^{(n)}) = \frac{2}{3} + \frac{1}{4} \left(\pi^{(n)} - \frac{2}{3} \right).$$

Regardless of the initial distribution, $\pi^{(n)} \rightarrow 2/3$ as $n \rightarrow \infty$. This is what we would like to see.

Example 4. Consider the transition matrix

$$\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

where the states are $\mathcal{X} = \{1, 2\}$. If $X_0 = 1$, then $X_t = 1$ for all even t and $X_t = 0$ for all odd t (and vice versa if $X_t = 1$). The distribution of X_t does not converge in distribution.

In subsequent sections, we will introduce concepts that allow us to state simple conditions ensuring that a Markov chain converges in distribution.

4. REDUCIBILITY

A chain is *irreducible* if every state can be reached from every other state.

To make this more precise, we say that state i *communicates* with state j , denoted $i \rightarrow j$, if there is positive probability that we can reach state j starting from state i , *i.e.* state i communicates with state j if

$$\mathbb{P}\{X_{n+m} = j | X_n = i\} > 0 \quad \text{for some } m < \infty.$$

Whether two states communicate can be determined by examining the transition matrix. if there exist states $i_1, \dots, i_{m-1}$ such that $p_{i,i_1}, p_{i_1,i_2}, \dots, p_{i_{m-1},j} > 0$, then $i \rightarrow j$, since

$$\mathbb{P}\{X_{n+1} = i_1, \dots, X_{n+m-1} = i_{m-1}, X_{n+m} = j | X_n = i\} = p_{i,i_1} p_{i_1,i_2} \cdots p_{i_{m-1},j} > 0.$$

In general, $i \rightarrow j$ iff $(\mathbf{P}^m)_{ij} > 0$ for some m .

Proposition 2. $\mathbb{P}\{X_{n+m} = j | X_n = i\} = (\mathbf{P}^m)_{ij}$.

Proof. For $m = 1$, $\mathbb{P}\{X_{n+1} = j | X_n = i\} = p_{ij} = (\mathbf{P})_{ij}$ by definition. By induction and the Markov property,

$$\begin{aligned} \mathbb{P}\{X_{n+m+1} = j | X_n = i\} &= \sum_{k \in \mathcal{X}} \mathbb{P}\{X_{n+m} = k | X_n = i\} \mathbb{P}\{X_{n+m+1} = j | X_{n+m} = k\} \\ &= \sum_{k \in \mathcal{X}} (\mathbf{P}^m)_{ik} p_{kj} = (\mathbf{P}^{m+1})_{ij}. \end{aligned}$$

□

If $i \rightarrow j$ and $j \rightarrow i$, then we say that states i and j *intercommunicate* and we write $i \leftrightarrow j$. A Markov chain is said to be *irreducible* if $i \leftrightarrow j$ for all $i, j \in \mathcal{X}$.

5. PERIODICITY

According to Feller, periodicity is a “nuisance” and “the beginner should concentrate his attention on chains without periodic states.” The Markov chains that you will encounter in MCMC are all aperiodic, but you should still understand what the assumption of aperiodicity is ruling out.

Example 5. Suppose we have a random walk on the state space $\mathcal{X} = \{0, 1, \dots, n-1\}$ arranged as an n -cycle (like a clock), where n is even. We start at state X_0 and at each period we move left or right one unit (each with probability $1/2$). Moving right from $n-1$ means moving to zero, while moving left from 0 means moving to $n-1$. The transition probabilities are

$$p_{ij} = \begin{cases} 1/2 & \text{if } j = i-1 \pmod n \text{ or } i+1 \pmod n \\ 0 & \text{otherwise} \end{cases}$$

If we are currently at an even numbered state, we will be at an odd numbered state next period and vice versa. Further, we can only return to the current state after an even number of periods.

If a chain starts in state, when will it return to that state? Let $T_i = \{t : (\mathbf{P}^t)_{ii} > 0\}$ be the number of periods for which there is a chance that the chain will return to state i . The *period* of state i is defined to be the greatest common divisor of T_i , i.e. the largest integer n such that n divides all $t \in T_i$. State i is said to be *aperiodic* if $\gcd(T_i) = 1$. A Markov chain is said to be *aperiodic* if all of its states are aperiodic: $\gcd(T_i) = 1$ for all $i \in \mathcal{X}$.

In an irreducible chain, all states have the same period.

Lemma 1. *If $\mathbf{P}$ is irreducible, then $\gcd(T_i) = \gcd(T_j)$ for all $i, j \in \mathcal{X}$.*

Proof. First, we show that all states have the same period for an irreducible Markov chain. Fix $i, j \in \mathcal{X}$. Since the chain is irreducible, there exist integers r and s such that $(\mathbf{P}^r)_{ij} > 0$ and $(\mathbf{P}^s)_{ji} > 0$. Let $m = r + s$ so that $m \in T(i) \cap T(j)$ (since it's feasible to go from i to j and back to i in $m = r + s$ steps, and similarly to go from j to i and back to j). Further, $T(i) \subset \{k : k = t - m \text{ for } t \in T(j), t > m\}$ (if you can return to i in k steps, you must be able to return to j in $k + m$ steps, though there may also be shorter paths). Thus, $\gcd(T_j)$ divides all elements of T_j , including m , so it must also divide all elements of T_i . This proves that $\gcd(T_j) \leq \gcd(T_i)$. Reversing the roles of i and j proves that $\gcd(T_i) \leq \gcd(T_j)$. Hence, $\gcd(T_i) = \gcd(T_j)$. $\square$

The important result is that all states in an irreducible and aperiodic chain intercommunicate after a finite number of periods. The proof is somewhat technical and is omitted.

Proposition 3. *If $\mathbf{P}$ is irreducible and aperiodic, then there exists an integer n such that*

$$(\mathbf{P}^n)_{ij} > 0$$

for all $i, j \in \mathcal{X}$.

Proof. See Levin *et al.*, p. 8. $\square$

6. STATIONARY DISTRIBUTIONS

A probability distribution π on $\mathcal{X}$ satisfying

$$\pi \mathbf{P} = \pi$$

is called a *stationary* (also *invariant* or *equilibrium*) distribution of the Markov Chain. Clearly, if the chain somehow reaches the stationary distribution, it stays there (in the sense that if $\pi^{(n)} = \pi$ for any n , then $\pi^{(m)} = \pi$ for all $m \geq n$). In this section, we consider (i) the existence of stationary distributions, (ii) their uniqueness, and (iii) whether a chain will converge to its stationary distribution.

Spoiler alert: every irreducible, aperiodic Markov chain has a unique stationary distribution π and $\pi^{(n)} \rightarrow \pi$.

6.1. Some Preliminaries. Before proving the main results, we require a few preliminary results. For $i \in \mathcal{X}$, define the *first return time* for i to be

$$\tau_i = \min\{t \geq 1 : X_t = i\}.$$

If the chain never reaches state i , set $\tau_i = \infty$. We will show below that this occurs with probability zero. We will also show that the expected return time is finite, regardless of which state the chain starts in.

To simplify the notation, we will let $\mathbb{P}_i$ and $\mathbb{E}_i$ denote probabilities and expectations, respectively, which are conditional upon $X_0 = i$. For example, $\mathbb{P}_i\{X_t = j\} = \mathbb{P}\{X_t = j | X_0 = i\}$.

Proposition 4. *In an irreducible and aperiodic Markov chain, $\mathbb{P}_i\{\tau_j < \infty\} = 1$ and $\mathbb{E}_i(\tau_j) < \infty$ for all $i, j \in \mathcal{X}$.*

Proof. By Proposition 3, there exists $n < \infty$ such that

$$\mathbb{P}_i\{X_t + n = j\} = (\mathbf{P}^n)_{ij} > 0$$

for all $i, j \in \mathcal{X}$. Let

$$\epsilon = \min_{i, j \in \mathcal{X}} \mathbb{P}_i\{X_n = j\}.$$

The probability of a return time greater than n is (for any i and j)

$$\mathbb{P}_i\{\tau_j > n\} \leq \mathbb{P}_i\{X_n \neq j\} \leq 1 - \epsilon.$$

Regardless of which state was reached at time n , the conditional probability of not returning to state j in an additional n periods is also less than $1 - \epsilon$ so

$$\mathbb{P}_i\{\tau_j > 2n\} = \mathbb{P}_i\{\tau_j > n\} \mathbb{P}\{\tau_j > 2n | X_0 = i, \tau_j > n\} < (1 - \epsilon)^2.$$

Iterating this argument shows that

$$\mathbb{P}_i\{\tau_j > rn\} \leq (1 - \epsilon)^r \quad \text{for } r = 1, 2, \dots$$

Taking $r \rightarrow \infty$ shows that $\mathbb{P}_i\{\tau_j = \infty\} = 0$.

Next, we calculate $\mathbb{E}_i(\tau_j)$. Since τ_j takes only positive integral values, we have the following well-known result,

$$\begin{aligned} \mathbb{E}_i(\tau_j) &= \sum_{t=1}^{\infty} t \mathbb{P}_i\{\tau_j = t\} = \sum_{t=1}^{\infty} \sum_{s=1}^t \mathbb{P}_i\{\tau_j = t\} \\ &= \sum_{s=1}^{\infty} \sum_{t=s}^{\infty} \mathbb{P}_i\{\tau_j = t\} = \sum_{s=1}^{\infty} \mathbb{P}_i\{\tau_j \geq s\} = \sum_{s=0}^{\infty} \mathbb{P}_i\{\tau_j > s\} \end{aligned}$$

Divide the last sum into groups of n terms and use the bound $\mathbb{P}_i\{\tau_j > rn\} \leq (1 - \epsilon)^r$ derived above to obtain

$$\begin{aligned} \mathbb{E}_i(\tau_j) &= \sum_{r=0}^{\infty} \sum_{k=rn}^{(r+1)n-1} \mathbb{P}_i\{\tau_j > k\} \leq \sum_{r=0}^{\infty} \sum_{k=rn}^{(r+1)n-1} \mathbb{P}_i\{\tau_j > rn\} \\ &= n \sum_{r=0}^{\infty} \mathbb{P}_i\{\tau_j > rn\} \leq n \sum_{r=0}^{\infty} (1 - \epsilon)^r = \frac{n}{\epsilon} < \infty. \end{aligned}$$

□

6.2. Existence of Stationary Distributions.

Proposition 5. *For any irreducible and aperiodic Markov chain, there exists at least one stationary distribution.*

Proof. Let $\tilde{\pi}_j$ denote the expected number of visits to state j before returning to state i ,

$$\tilde{\pi}_j = \sum_{t=0}^{\infty} \mathbb{P}_i\{X_t = j, \tau_i > t\}.$$

From Proposition 4, $\tilde{\pi}_j \leq \mathbb{E}_i(\tau_i) < \infty$ for all $i, j \in \mathcal{X}$.

Next, we propose

$$\pi_j = \frac{\tilde{\pi}_j}{\sum_{\ell \in \mathcal{X}} \tilde{\pi}_\ell} \quad (j \in \mathcal{X})$$

as the stationary probabilities. These are clearly non-negative and sum to one, so we only need to verify that this choice of π is, in fact, stationary. We can defer normalizing the $\tilde{\pi}_j$'s until later, so it is enough to show

$$\tilde{\pi}_j = \sum_{k \in \mathcal{X}} \tilde{\pi}_k p_{kj}$$

for $j \in \mathcal{X}$. By definition,

$$\sum_{k \in \mathcal{X}} \tilde{\pi}_k p_{kj} = \sum_{k \in \mathcal{X}} \sum_{t=0}^{\infty} \mathbb{P}_i\{X_t = k, \tau_i > t\} p_{kj}.$$

Since the event $\{\tau_i > t\}$ is determined by $X_0, \dots, X_t$, the Markov property implies that

$$\mathbb{P}_i\{X_t = k, X_{t+1} = j, \tau_i > t\} = \mathbb{P}_i\{X_t = k, \tau_i > t\} p_{kj}.$$

Substituting this into the previous expression, reversing the order of summation, and rearranging a bit gives

$$\begin{aligned} \sum_{k \in \mathcal{X}} \tilde{\pi}_k p_{kj} &= \sum_{k \in \mathcal{X}} \sum_{t=0}^{\infty} \mathbb{P}_i\{X_t = k, X_{t+1} = j, \tau_i > t\} \\ &= \sum_{t=0}^{\infty} \sum_{k \in \mathcal{X}} \mathbb{P}_i\{X_t = k, X_{t+1} = j, \tau_i > t\} \\ &= \sum_{t=0}^{\infty} \mathbb{P}_i\{X_{t+1} = j, \tau_i > t\} = \sum_{t=1}^{\infty} \mathbb{P}_i\{X_t = j, \tau_i \geq t\} \\ &= \sum_{t=0}^{\infty} \mathbb{P}_i\{X_t = j, \tau_i \geq t\} - \mathbb{P}_i\{X_0 = j, \tau_i \geq 0\} \\ &= \sum_{t=0}^{\infty} \mathbb{P}_i\{X_t = j, \tau_i > t\} + \sum_{t=0}^{\infty} \mathbb{P}_i\{X_t = j, \tau_i = t\} - \mathbb{P}_i\{X_0 = j\}. \end{aligned}$$

The first term on the last line is $\tilde{\pi}_j$, so we need to show that the last two terms sum to zero. We treat the cases $i = j$ and $i \neq j$ separately. If $i = j$, $\mathbb{P}_i\{X_0 = j\} = \mathbb{P}_i\{X_0 = i\} = 1$ and

$$\sum_{t=0}^{\infty} \mathbb{P}_i\{X_t = j, \tau_i = t\} = \mathbb{P}_i\{X_{\tau_i} = i\} = 1,$$

so the last two terms cancel. If $i \neq j$, the last two terms are both zero: $\mathbb{P}_i\{X_0 = j\} = \mathbb{P}_i\{X_{\tau_i} = j\} = 0$. This proves that $\pi = \pi P$. $\square$

Remark 1. In the proof, $\tilde{\pi}_j$ was the expected number of visits to state j before returning to the starting state i . Since the return time equals the sum of the number of visits to states other than i before the first return, plus the return visit to state i , we have $\sum_{j \in \mathcal{X}} \tilde{\pi}_j = \mathbb{E}_i(\tau_i)$ and

$$\pi_i = \frac{\tilde{\pi}_i}{\mathbb{E}_i(\tau_i)} = \frac{1}{\mathbb{E}_i(\tau_i)}.$$

That is, the stationary probability of being in state i is proportional to the proportion of time spent in that state.

6.3. Uniqueness of the Invariant Distribution.

Lemma 2. If P is irreducible and $\mathbf{h}$ is a right eigenvector of P , then all elements of $\mathbf{h}$ are equal.

Proof. Suppose $P\mathbf{h} = \mathbf{h}$. Let $M = \max_{i \in \mathcal{X}} h_i$. Since $\mathcal{X}$ is finite, there exists $i_0 \in \mathcal{X}$ such that $h_{i_0} = M$ and

$$h_{i_0} = \sum_{j \in \mathcal{X}} p_{i_0,j} h_j.$$

If $p_{i_0,j} > 0$, then $h_j = M$, for if this were not the case we would have

$$M = p_{i_0,j} h_j + \sum_{k \neq j} p_{i_0,k} h_k < M \left(p_{i_0,j} + \sum_{k \neq j} p_{i_0,k} \right) = M,$$

a contradiction. Therefore, $h_j = M$ if $p_{i_0,j} > 0$. Since P is irreducible, there exists a sequence $i_0, i_1, \dots, i_n = j$ with $p_{i_0,i_1} > 0, \dots, p_{i_{n-1},i_n} > 0$ for which $M = h_{i_0} = h_{i_1} = \dots = h_{i_n}$ so $h_j = M$ for all $j \in \mathcal{X}$. $\square$

Proposition 6. If the transition matrix P for a Markov chain is irreducible and aperiodic, it has a unique stationary distribution.

Proof. By Proposition 5, there exists at least one stationary distribution for P . By the preceding lemma, $\dim(P - I) = 1$ (since $(P - I)\mathbf{h} = \mathbf{0}$ implies that $\mathbf{h} = \lambda(1, \dots, 1)^T$). Since the row and column rank of $P - I$ are the same, $\dim(P^T - I) = 1$ as well. This implies that there is a unique row vector π satisfying $\pi(P - I) = \mathbf{0}$ with $\sum_{i \in \mathcal{X}} \pi_i = 1$. $\square$

6.4. The Markov Convergence Theorem. We reach (finally) the central result.

Theorem 1 (Markov). Let $\{X_t\}$ be an irreducible aperiodic Markov chain on a finite state space $\mathcal{X}$ with transition matrix P and any initial distribution $\pi^{(0)}$ on $\mathcal{X}$. Then, there exists a unique stationary distribution π such that

$$\lim_{n \rightarrow \infty} \pi^{(n)} = \pi.$$

Remark 2. Let $\mathbb{P}_{\pi^{(n)}}(A) = \sum_{i \in A} \pi_i^{(0)}$ denote the probability measure associated with $\pi^{(n)}$ and similarly for $\mathbb{P}_{\pi}$. Because $\mathcal{X}$ (and hence A) is finite, the convergence is uniform over all events $A \subset \mathcal{X}$, i.e.

$$\lim_{n \rightarrow \infty} \sup_{A \subset \mathcal{X}} |\mathbb{P}_{\pi^{(n)}}(A) - \mathbb{P}_{\pi}(A)| = 0.$$

The same result extends to general state spaces $\mathcal{X}$, where this is not so trivial.

Proof. We sketch a proof using the method of coupling: we construct an independent Markov chain $\{X'_t\}$ with the same transition matrix $\mathbf{P}$ but with initial distribution equal to the stationary distribution π . We show that the two chains $\{X_t\}$ and $\{X'_t\}$ eventually “meet” (i.e., $X_\tau = X'_\tau$ for some $\tau < \infty$ with probability one). We then construct a third Markov chain $\{X''_t\}$ with $X''_t = X_t$ for $t \leq \tau$ and subsequently $\{X''_t\}$ has the same distribution as $\{X'_t\}$.

Given an initial distribution $\pi^{(0)}$ on $\mathcal{X}$ and a transition matrix $\mathbf{P}$, we can construct a Markov chain from an i.i.d. sequence $U_0, U_1, \dots$ of $\text{Uniform}(0, 1)$ random variables using the inverse c.d.f. method (see, for example, Grimmett and Stirzaker, *Probability and Random Processes*, 3rd ed., pp. 122-123). This amounts to $X'_0 = \phi(U_0)$ and $X'_{n+1} = \phi(X'_n, U'_{n+1})$, where the sequence $\{U'_t\}$ is independent of $\{X_t\}$.

Details may be found in Levin *et al.*, pp. 73-74. □